

1.SOLVE ME FIRST

```
import java.util.Scanner;

public class Solution
{

    public static void main(String[] args)
    {
        Scanner id = new Scanner(System.in);
        int a = id.nextInt();
        int b = id.nextInt();
        int sum = a + b;
        System.out.println(sum);
    }
}
```

2.JAVA IF-ELSE (WEIRD OR NOT)

```
import java.util.*;

public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        if(n%2==1)
            System.out.println("Weird");
        else
        {
            if(n>=2 && n<=5)
                System.out.println("Not Weird");
            else if(n>=6 && n<=20)
                System.out.println("Weird");
            else
                System.out.println("Not Weird");
        }
    }
}
```

3.GRADING STUDENTS

SOLUTION-1:

```
import java.util.*;

public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for(int i=1;i<=n;i++)
        {
            int m = sc.nextInt();
            if ( (m>=38) && (m%5>=3) )
            {
                System.out.println(( (m/5) *5)+5);
            }
            else
                System.out.println(m);
        }
    }
}
```

SOLUTION-2:

```
import java.util.Scanner;

public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for(int i=1;i<=n;i++)
        {
            int m = sc.nextInt();
            if (m>=38)
            {
                if(m%5==4)
                    System.out.println(m+1);
                else if(m%5==3)
                    System.out.println(m+2);
                else
                    System.out.println(m);
            }
            else
                System.out.println(m);
        }
    }
}
```

4.A VERY BIG SUM

```
import java.util.Scanner;

public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        long sum = 0;
        for(int i=1; i<=n; i++)
        {
            sum = sum + sc.nextLong();
        }
        System.out.println(sum);
    }
}
```

5.PLUS MINUS

```
import java.util.*;

public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[]=new int[n];
        int c1 = 0; // To store +ve count
        int c2 = 0; // To store +ve count
        int c3 = 0; // To store Zeroes count
        for(int i=0; i<n; i++)
        {
            arr[i]=sc.nextInt();
            if(arr[i]>0)
                c1++;
            else if(arr[i]<0)
                c2++;
            else
                c3++;
        }
        System.out.println((float)c1/n);
        System.out.println((float)c2/n);
        System.out.println((float)c3/n);
    }
}
```

6.MINI-MAX SUM

SOLUTION-1

```
import java.util.*;

public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        long arr[]=new long[5];
        long sum =0;
        long max = 0;
        long min = Long.MAX_VALUE;
        for(int i=0;i<5;i++)
        {
            arr[i]=sc.nextInt();
            sum += arr[i];
            // To find max element
            if(arr[i]>max)
                max = arr[i];
            //To find min element
            if(arr[i]<min)
                min = arr[i];

        }
        System.out.print(sum-max+" ");
        System.out.print(sum-min);
    }
}
```

SOLUTION-2:

```
import java.util.*;

public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        long arr[]=new long[5];
        long sum =0;

        for(int i=0;i<5;i++)
        {
            arr[i]=sc.nextInt();
            sum += arr[i];
        }
        Arrays.sort(arr);
        System.out.print(sum-arr[4]+" ");
        System.out.print(sum-arr[0]);
    }
}
```

7.BIRTHDAY CAKE CANDLES

```
import java.util.Scanner;
public class Solution
{
    public static void main(String args[])
    {
        Scanner obj=new Scanner(System.in);
        int n=obj.nextInt();
        int a[]=new int[n];
        int max = 0;
        int count = 0;
        // 5 2 3 5 6 6 7
        for(int i=0;i<n;i++)
        {
            a[i]=obj.nextInt();
            if(a[i]>=max){
                count = (a[i]==max)?count+1:1;
                max = a[i];
            }
        }

        System.out.print(count);
    }
}
```

8.COMPARE THE TRIPLETS

SOLUTION-1:

```
import java.util.*;
public class Solution
{
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);
        int a[]=new int[3];
        int b[]=new int[3];
        int player1 = 0;
        int player2 = 0;
        for(int i=0;i<3;i++)
        {
            if(i==0)
            {
                a[i]=sc.nextInt();
                a[i+1]=sc.nextInt();
                a[i+2]=sc.nextInt();
            }
            else
            {
                b[i-1]=sc.nextInt();
                b[i]=sc.nextInt();
                b[i+1]=sc.nextInt();
            }
        }
        for(int i=0;i<3;i++)
        {
            if(a[i]>b[i])
                player1++;
            else if(a[i]<b[i])
                player2++;
        }
        System.out.println(player1+" "+player2);
    }
}
```

SOLUTION-2:

```
import java.util.*;
public class Solution
{
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);
        int a[]=new int[3];
        int b[]=new int[3];
        int player1 = 0;
        int player2 = 0;
        for(int i=0;i<3;i++)
        {
            a[i]=sc.nextInt();
        }
        for(int i=0;i<3;i++)
        {
            b[i]=sc.nextInt();
        }
        for(int i=0;i<3;i++)
        {
            if(a[i]>b[i])
                player1++;
            else if(a[i]<b[i])
                player2++;
        }
        System.out.println(player1+" "+player2);
    }
}
```

9.DIAGONAL DIFFERENCE

```
import java.util.*;
public class Solution
{
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[][]=new int[n][n];
        int ld =0;
        int rd = 0;
        for(int i=0;i<n;i++)//rows
        {
            for(int j=0;j<n;j++)//columns
            {
                arr[i][j]=sc.nextInt();
                if(i==j)//left diagonal
                    ld = ld+ arr[i][j];
                if(i+j == (n-1))//right diagonal
                    rd = rd+arr[i][j];
            }
        }
        System.out.println(Math.abs(ld-rd));
    }
}
```

10.BREAKING THE RECORDS

```
import java.util.*;
public class Solution
{
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[]=new int[n];
        for(int i=0;i<n;arr[i++]=sc.nextInt());
        int min = arr[0];
        int max = arr[0];
        int best = 0,worst = 0;
        for(int i=0;i<n;i++)
        {
            if(arr[i]>max)
            {
                max = arr[i];
                best++;
            }
            if(arr[i]<min)
            {
                min = arr[i];
                worst++;
            }
        }
        System.out.println(best+" "+worst);
    }
}
```

11.DIVISIBLE SUM PAIRS

```
import java.util.*;
public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int k = sc.nextInt();
        int arr[]=new int[n];
        for(int i=0;i<n;i++)
        {
            arr[i]=sc.nextInt();
        }
        int pair = 0;
        for(int i=0;i<n;i++)
        {
            for(int j=i+1;j<n;j++)
            {
                int sum = arr[i]+arr[j];
                if(sum%k==0)
                    pair++;
            }
        }
        System.out.println(pair);
    }
}
```

12.SALES BY MATCH

```
import java.util.*;
public class Solution
{
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[]=new int[n];
        for(int i=0;i<n;i++)
            arr[i]=sc.nextInt();
        int pair = 0;
        for(int i=0;i<n;i++)
        {
            int count = 1;
            for(int j=i+1;j<n;j++)
            {
                if(arr[i]==arr[j] && arr[i]!=0)
                {
                    count++;
                    arr[j]=0;
                }
            }
            pair = pair + (count/2);
        }
        System.out.println(pair);
    }
}
```

13.SAVE THE PRISONER!

```
import java.util.*;

public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int x = sc.nextInt();
        for(int i=0;i<x;i++)
        {
            int n = sc.nextInt();
            int sweets = sc.nextInt();
            int start = sc.nextInt();
            sweets %= n;
            int last = (start+sweets-1)%n;
            if(last==0)
                System.out.println(n);
            else
                System.out.println(last);
        }
    }
}
```

14.SHERLOCK AND SQUARES

```
import java.util.*;
public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int q = sc.nextInt();
        for(int i=0;i<q;i++)
        {
            int start = sc.nextInt();
            int end = sc.nextInt();
            start = (int)Math.ceil(Math.sqrt(start));
            end = (int)Math.floor(Math.sqrt(end));
            System.out.println(end-start+1);
        }
    }
}
```

15.DAY OF THE PROGRAMMER

```
import java.util.*;

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int y = sc.nextInt();
        int date = 13;
        if(y < 1918){
            if(y%4==0)
                date = 12;
        }
        else if(y>1918){
            if(y%400 == 0 || (y%4==0 && y%100!=0))
                date = 12;
        }
        else {
            date = 26;
        }
        String result = date + ".09."+y;
        System.out.println(result);
    }
}
```

16.BETWEEN TWO SETS

```
import java.util.*;
public class Solution
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int l1 = sc.nextInt();
        int l2 = sc.nextInt();
        int a[]=new int[l1];
        int b[]=new int[l2];
        //Input for first array
        for(int i=0;i<l1;i++)
            a[i]=sc.nextInt();
        // Input for second array
        for(int i=0;i<l2;i++)
            b[i]=sc.nextInt();

        int count = 0;
        for(int i=1;i<=100;i++)
        {
            boolean multiple = true;
            for(int j=0;j<l1;j++)
            {
                if(i%a[j]!=0)
                {
                    multiple = false;
                    break;
                }
            }
            if(multiple == true)
            {
                boolean factor = true;
                for(int k=0;k<l2;k++)
                {
                    if(b[k]%i!=0)
                    {
                        factor = false;
                        break;
                    }
                }
                if(factor == true)
                    count++;
            }
        }
    }
}
```

```
        System.out.println(count);  
    }  
}
```